

Emerging Career Opportunities

Embedded technology is one of the fastest growing sectors in India, and is likely to remain so for a long time. The proliferation of digital devices and equipment across various industry verticals such as automotives, telecom, consumer electronics, etc. is pushing the need for embedded systems engineers and professionals.

Semiconductor-based products such as VLSI chips are used in the development and design of hardware for embedded systems; the semiconductor and embedded design industries are closely connected.

"India is playing an important role in the progress of the global semiconductors and embedded technology-driven industry. A number of factors such as a rapidly growing domestic market, a strong education system, short product lead-times, reduced entry barriers, government support, and improved infrastructure have resulted in the semiconductor and embedded systems industries emerging as the sunrise sectors in India," says Joe Lazar, director—Human Resources, Analog Devices Inc., which deals in embedded processing and Digital Signal Processing.

In this field, one can develop or design a completely new product, add features to or improve over the current line of products, or provide solutions to problems.

Needless to mention here, the demand for qualified people is greater than supply. There is the scope of building a bright future outside India as well.

Multinational companies that have R&D labs and design houses in India send their employees to their home countries for special training or on special projects.

A Choice Of Streams

As mentioned earlier, embedded systems are a combination of hardware and software, where developments take place in both areas separately—and have an impact on each other. Opportunities in terms of hardware development range from the design and development of simple products such as consumer and home appliances, to complex tasks such as automotive design, product debugging and testing, and more. On the software side, opportunities include embedded software development, working with RTOSes, debugging, and designing interfaces for embedded operating systems.



Candidates from an engineering background with a specialisation in the electronics, electrical engineering or computer science streams with clear fundamentals... are most suitable"

Vinodh Chelambathodi
General Manager—
Human Resources
Engineering and R & D
Services
HCL Technologies Ltd

There are two teams in organisations involved in the development or designing of embedded systems—hardware and software. Developing a program to be written onto the firmware/hardware is a challenging task: it involves working with RTOSes.

Embedded operating systems, too, are developed for certain specific products. Programmers need to have knowledge of the interface between different OSes, the hardware of the embedded devices, and the intricacies of the hardware while coding.

Programming and developing applications for embedded systems is considerably harder than that for personal computers. In the case of personal computers, the system can be given a reboot if there's an error in the code, but that's just not possible in the case of embedded systems (imagine just having to "restart" your washing machine if it goes dead!).

Teams of hardware engineers will have to understand and cope with issues, solutions, updates, and developments in software that may have an impact on them, and vice versa, software engineers need to have some hardware knowledge. Hardware engineers need basic knowledge of programming in C and assembly languages.

Essentially, hardware and software engineers work in sync with each other. There is an equal requirement for hardware as well as software engineers in the field of embedded systems and so one should select the stream of his/her choice—hardware or software.

The Basic Requirements

The key to starting off on an embedded systems career path is to pursue an education in engineering or computer science. The foundation for a successful career could be laid by acquiring a Bachelors or Masters engineering degree with a specialisation preferably in electronics, electrical engineering, or computer science. Aspirants from the telecommunications, industrial electronics and automation, instrumentation design, and VLSI design, too, have a chance to pursue a career in the field of embedded systems by opting for various diploma and post-graduate short-term training courses offered by institutes both government and private.

Some engineering colleges offer degree courses in embedded systems, while some offer embedded systems as an elective during the graduating year. Candidates must have knowledge of programming in C or C++, as well as hardware languages such as Verilog or VHDL. Also offered are courses for Masters or Research-level degrees in embedded systems and VLSI design. (For more on VLSI chip design, refer to *Chip Your Way Through, Digit*, September 2006).

The R&D labs and design houses of the companies that are into embedded systems usually look for engineers with a strong background in electronics, as such candidates tend to have the core knowledge and basic fundamentals of working with such things as circuit boards, microprocessors, and microcontrollers. Companies involved in software development for embedded devices look for candi-

Sectors For Career Opportunities

- ❑ Automobiles
- ❑ Aviation/Aerospace
- ❑ Consumer Electronics
- ❑ Industrial Automations And Controls
- ❑ Computers and peripherals
- ❑ IT/ITES
- ❑ Medical Instruments
- ❑ Military
- ❑ Semiconductors
- ❑ Telecommunication